# Camera calibration (CSI)


**Camera calibration (CSI)**

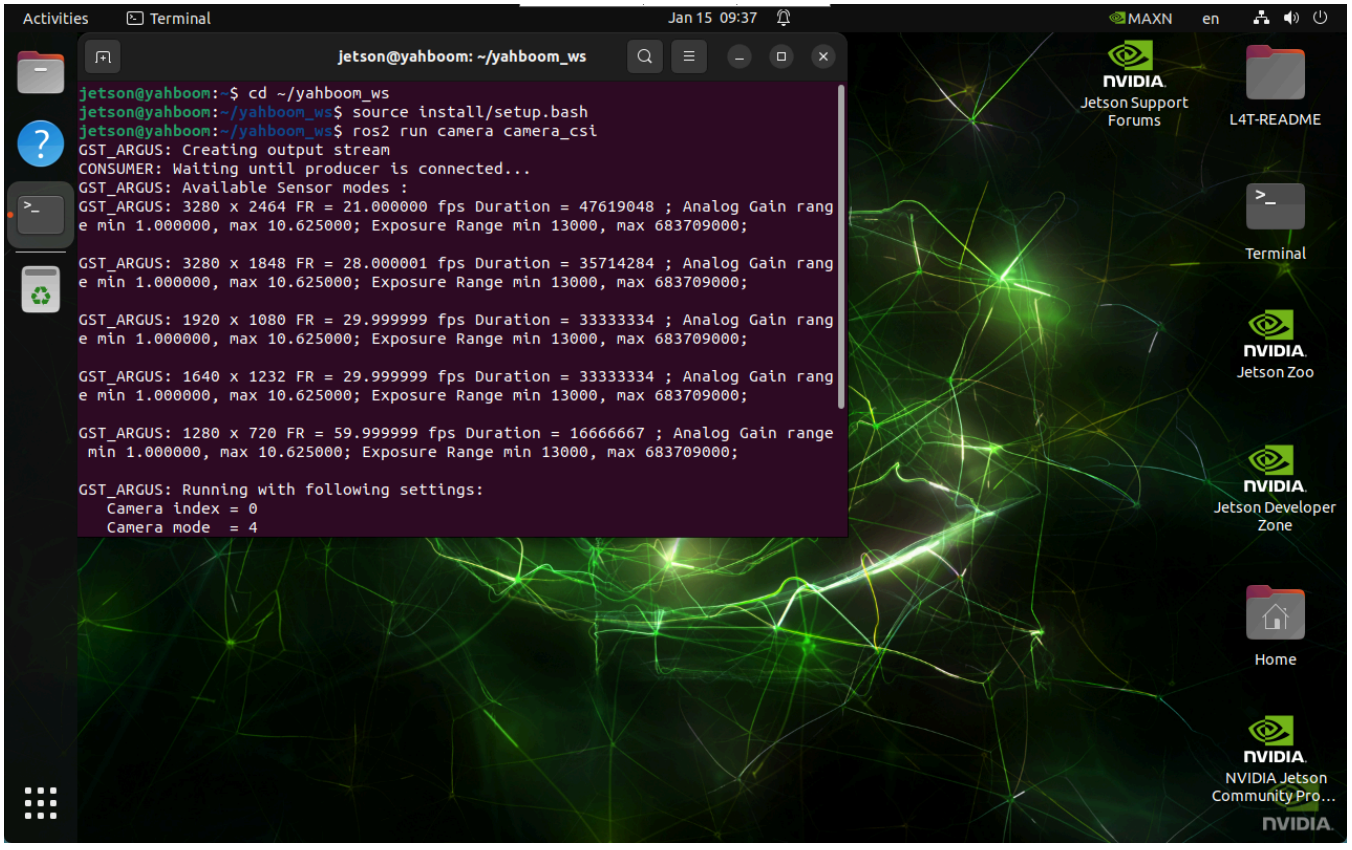
1. Start the camera
2. Image calibration
    - 2.1. Calibration board
    - 2.2. View topics
    - 2.3. Image calibration
        - 2.3.1. Install the image calibration tool
        - 2.3.2. Start the image calibration tool
        - 2.3.3. Calibration process
        - 2.3.4. Calibration results


## 1. Start the camera

```
cd ~/yahboom_ws
```

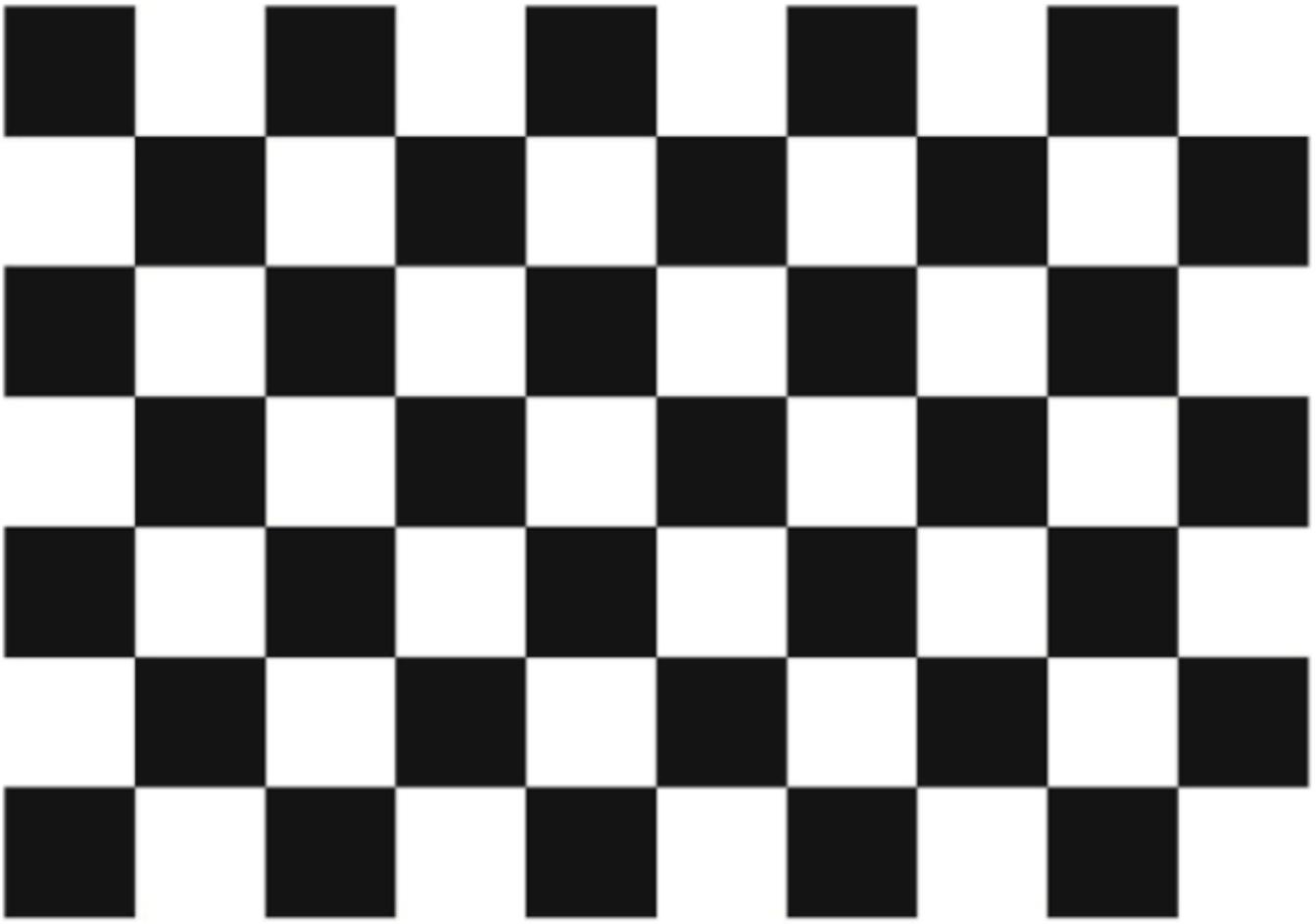
```
source install/setup.bash
```

```
ros2 run camera camera_csi
```



# 2. Image calibration

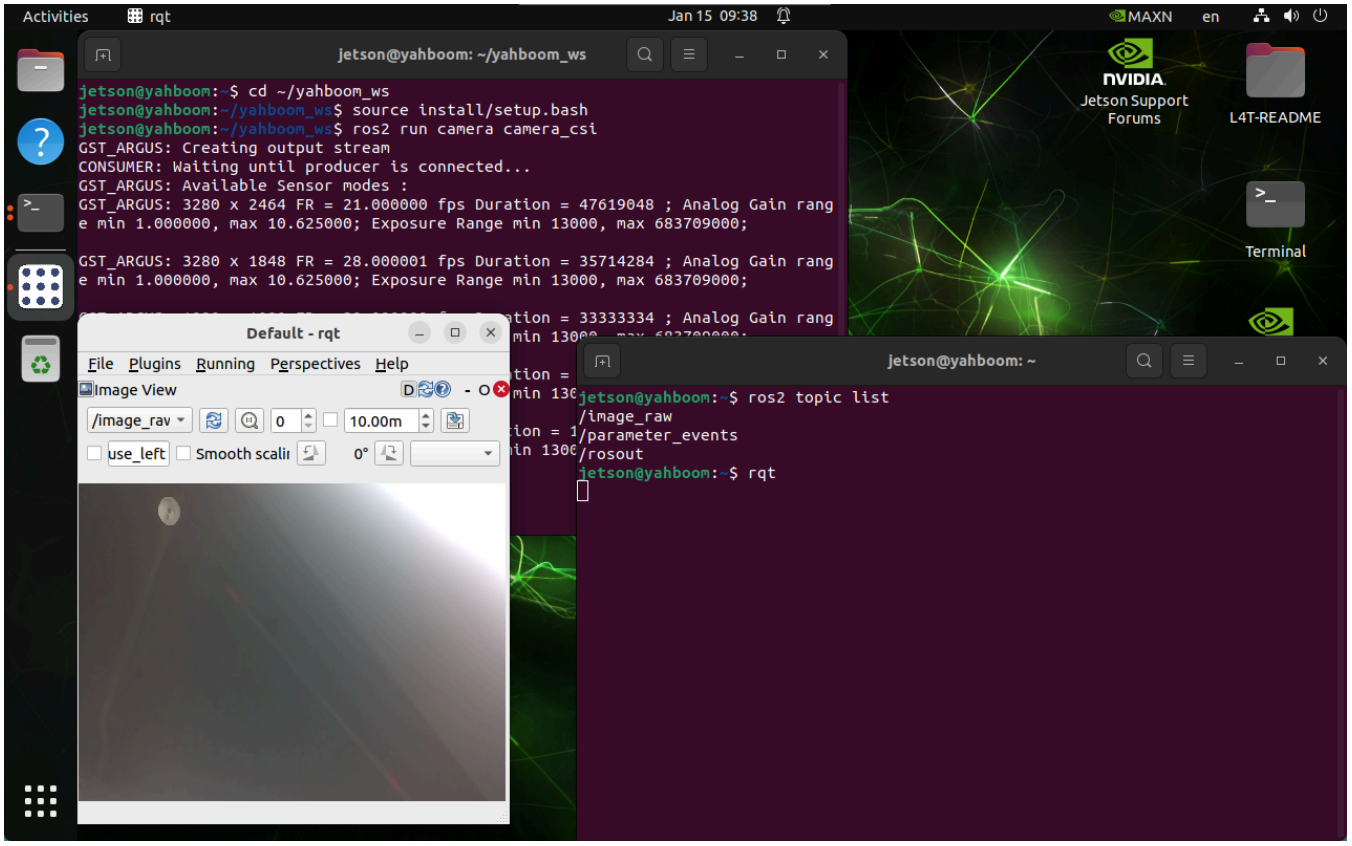
## 2.1. Calibration board



7×10 | Size: 20mm

## 2.2. View topics

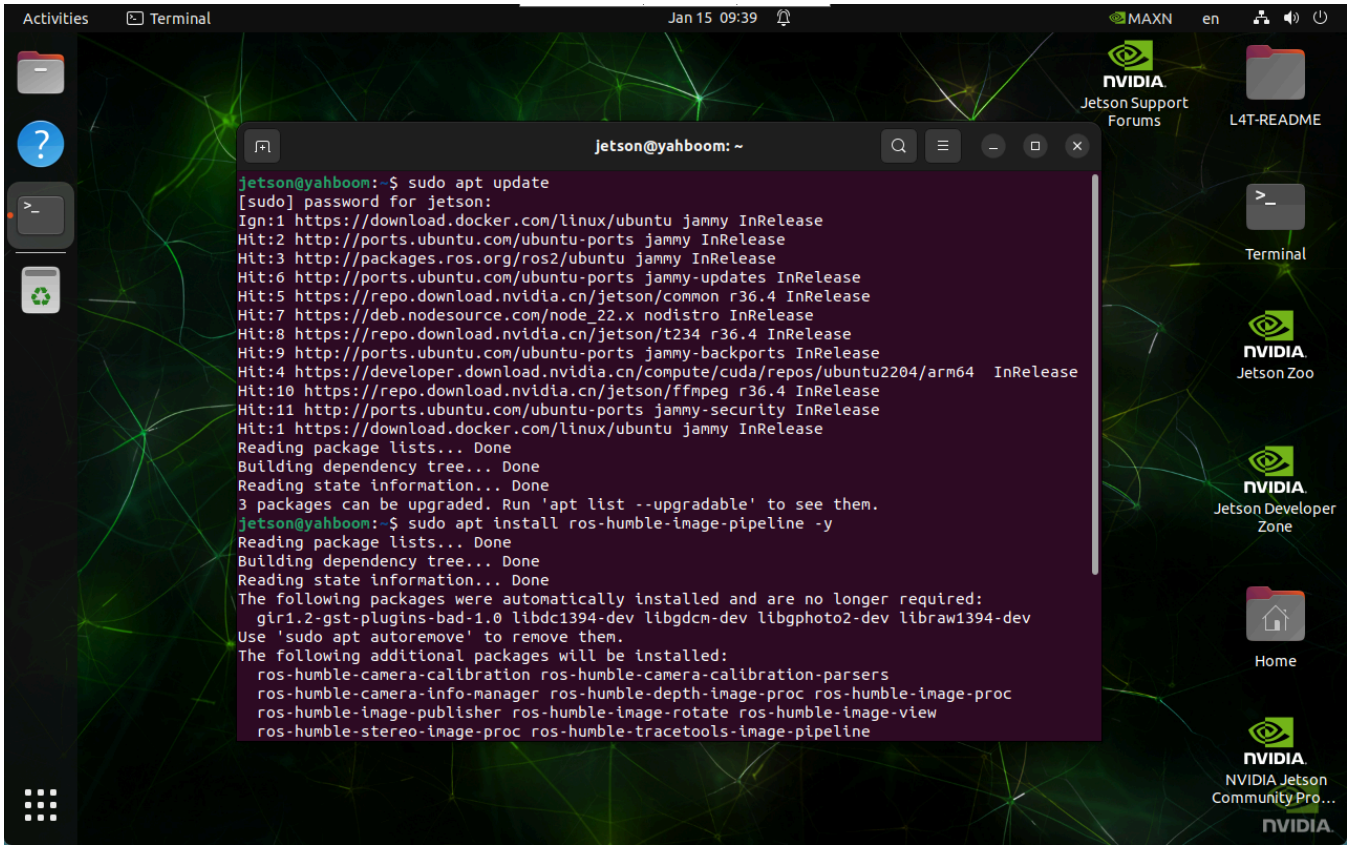
```
ros2 topic list
```




## 2.3. Image calibration

### 2.3.1. Install the image calibration tool

```
sudo apt update
sudo apt install ros-${ROS_DISTRO}-image-pipeline -y
```



## 2.3.2. Start the image calibration tool

Start the ROS2 camera calibration tool `camera_calibration` and perform the camera calibration: before calibrating, you need to start the camera first

```
ros2 run camera_calibration cameracalibrator --size 9x6 --square 0.02 --ros-args --remap /image:=/image_raw
```

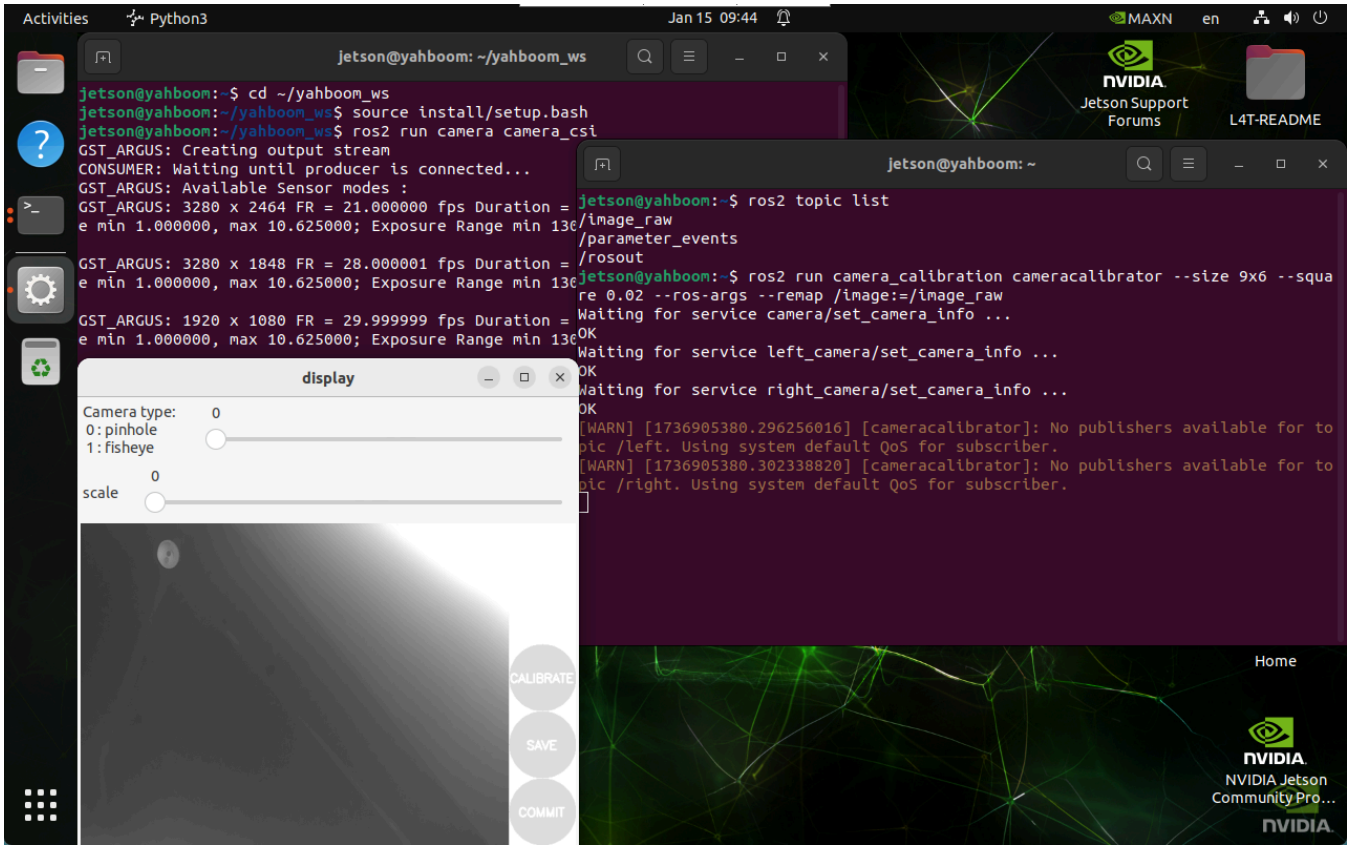
> **Parameter Description**

`--size`: the number of inner corner points of the chessboard → inner corner points of 9 rows and 6 columns

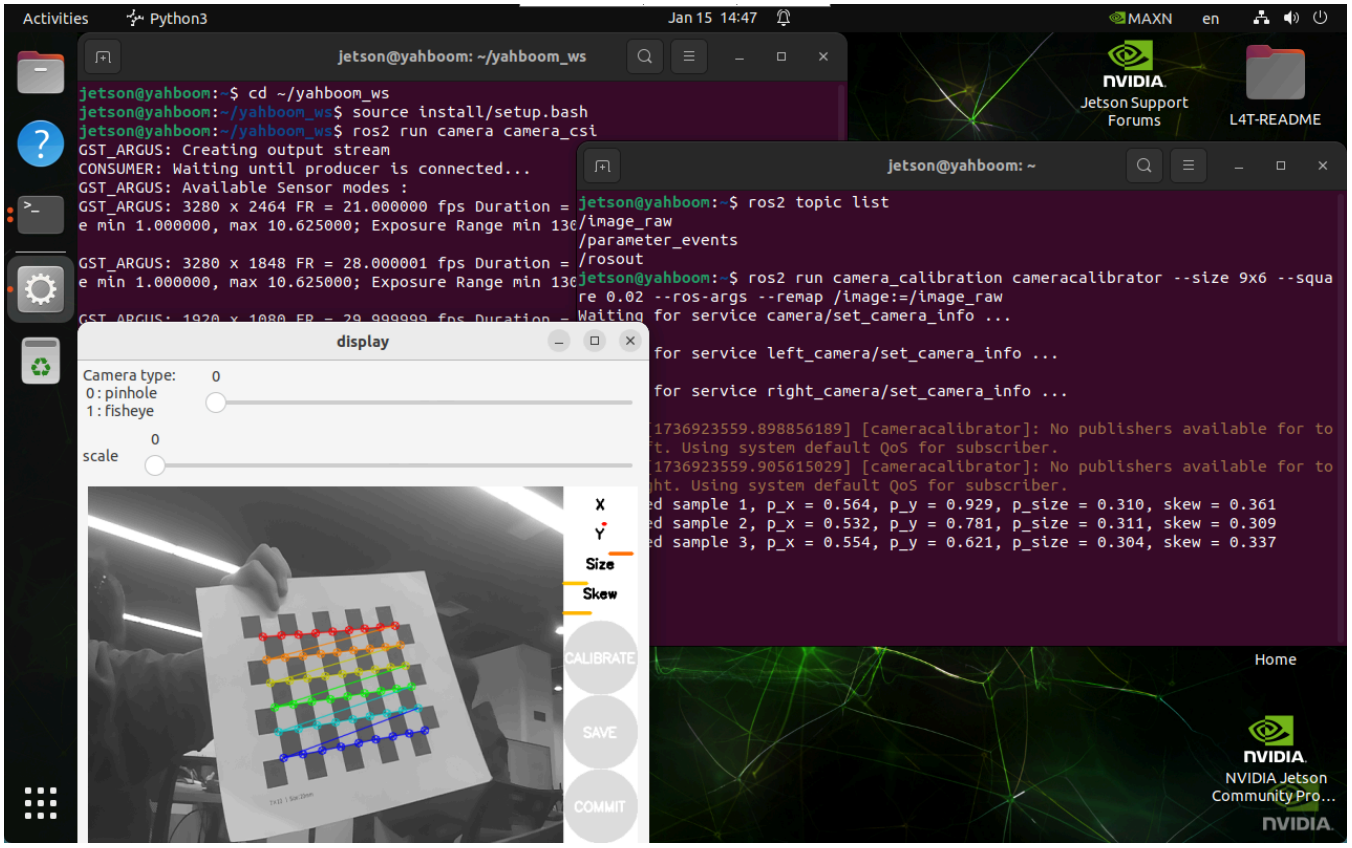
`--square`: the side length of the chessboard → 0.02 meters

`--ros-args`: pass ROS parameters

`--remap`: topic remapping → `/image_raw` as the image input stream



## 2.3.3. Calibration process




> **Parameter Description**

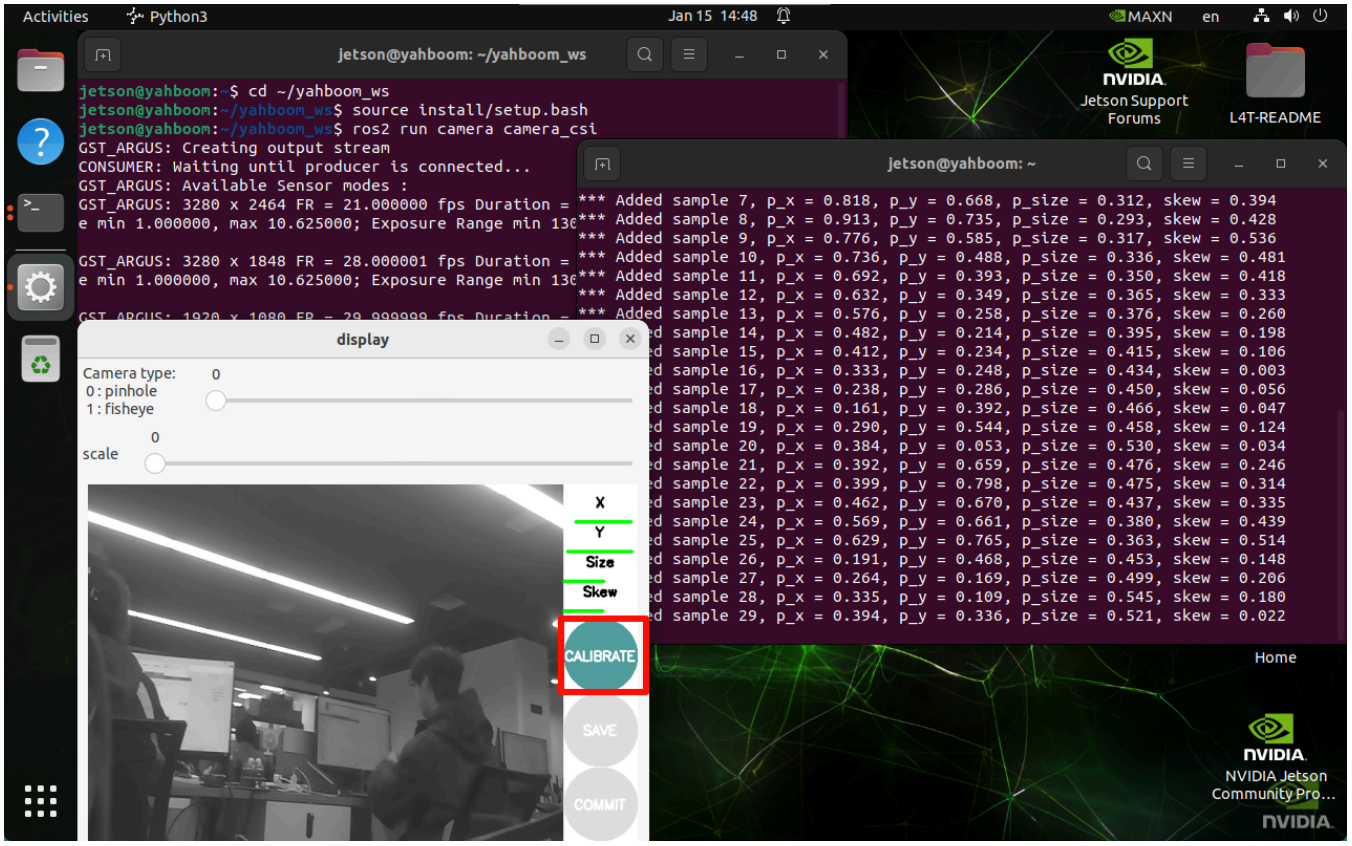
`X`: move the chessboard left and right within the camera's field of view

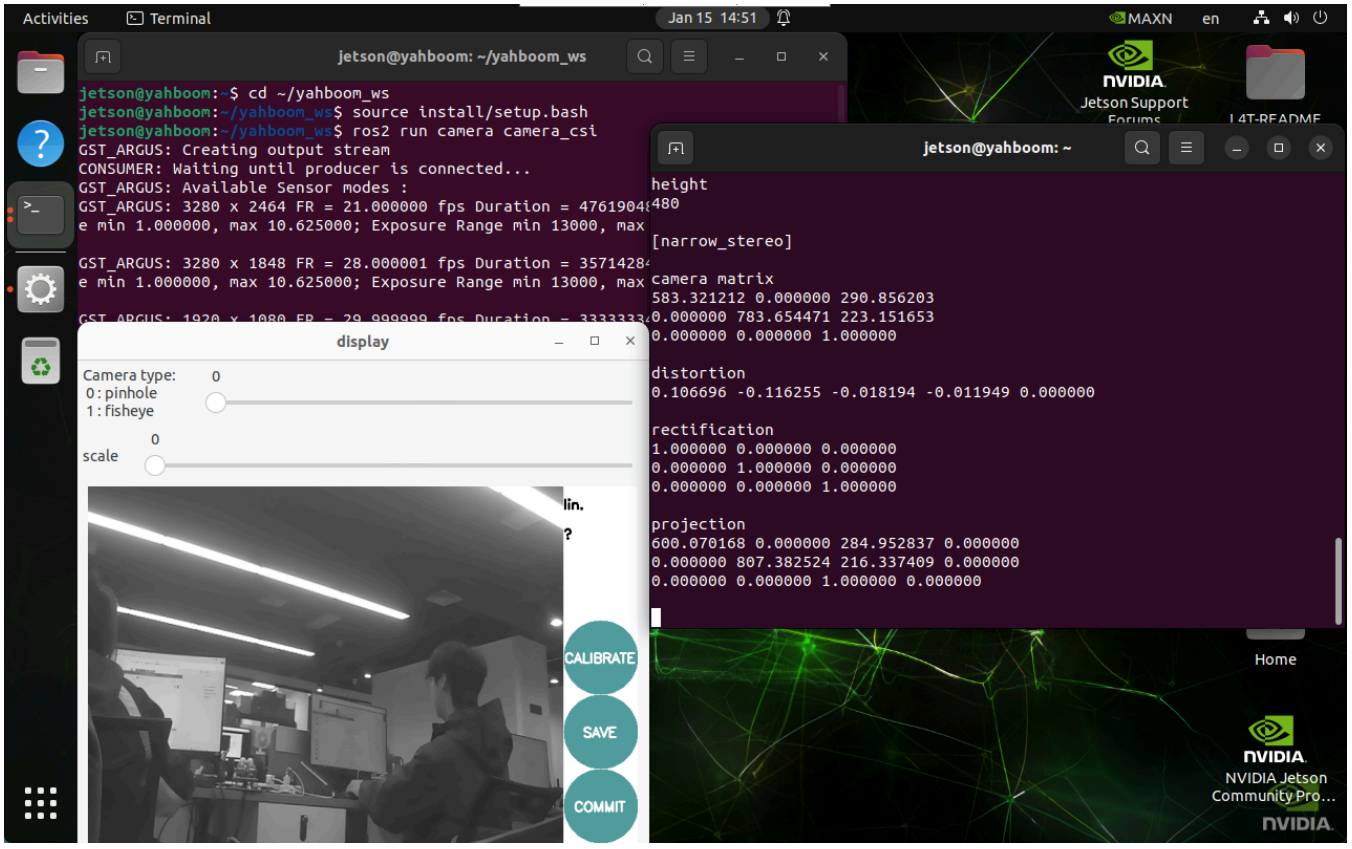
`Y`: move the chessboard up and down within the camera's field of view

`Size`: move the chessboard back and forth within the camera's field of view

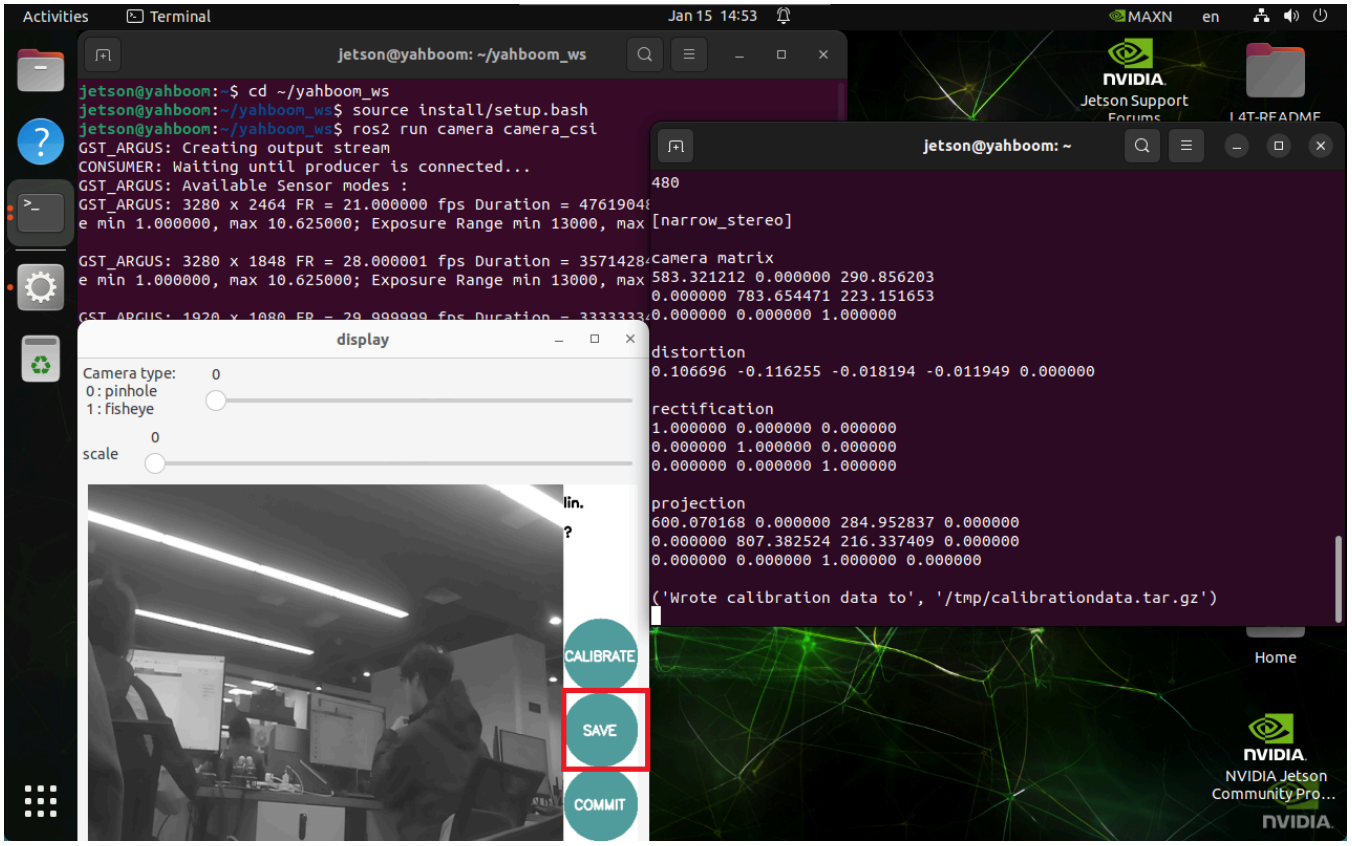
`Skew`: tilt and rotate the chessboard within the camera's field of view

When `X`, `Y`, `Size` and `Skew` turn green, you can click `CALIBRATE` to calculate the camera intrinsic parameters: the more images you calibrate, the longer it takes, and the program may get stuck




After the calibration is finished, click `SAVE` to save the results!




Close the program after saving successfully!

### 2.3.4. Calibration results

The default save location and file name of the calibration results:

```
/tmp/calibrationdata.tar.gz
```

Copy and decompress the calibration results:

```
sudo mv /tmp/calibrationdata.tar.gz ~
mkdir calibrationdata
tar -xzvf calibrationdata.tar.gz -C ~/calibrationdata
```

The decompressed files contain the calibration `*.png`, `ost.yaml` and `ost.txt` files. We mainly copy the data in the `ost.yaml` file into the `camera_usb.yaml` file.

> **ost.yaml**

Replace the `camera_usb.yaml` file data with the `image_width`, `image_height`, the `data` in `camera_matrix`, the `data` in `projection_matrix`, and the `data` in `distortion_coefficients`

```
image_width: 640
image_height: 480
camera_name: narrow_stereo
camera_matrix:
  rows: 3
  cols: 3
  data: [583.32121,   0.      , 290.8562 ,
```

```
          0.     , 783.65447, 223.15165,
          0.     ,   0.     ,   1.     ]
distortion_model: plumb_bob
distortion_coefficients:
  rows: 1
  cols: 5
  data: [0.106696, -0.116255, -0.018194, -0.011949, 0.000000]
rectification_matrix:
  rows: 3
  cols: 3
  data: [1., 0., 0.,
         0., 1., 0.,
         0., 0., 1.]
projection_matrix:
  rows: 3
  cols: 4
  data: [600.07017,   0.     , 284.95284,   0.     ,
          0.     , 807.38252, 216.33741,   0.     ,
          0.     ,   0.     ,   1.     ,   0.     ]
```

> **camera_csi.yaml**

The modified file:

```
%YAML:1.0
---
image_width: 1920
image_height: 1080
camera_name: camera
camera_matrix: !!opencv-matrix
  rows: 3
  cols: 3
  dt: d
  data: [583.32121,   0.     , 290.8562 ,
          0.     , 783.65447, 223.15165,
          0.     ,   0.     ,   1.     ]
distortion_model: plumb_bob
distortion_coefficients: !!opencv-matrix
  rows: 1
  cols: 5
  dt: d
  data: [0.106696, -0.116255, -0.018194, -0.011949, 0.000000]
rectification_matrix:
  rows: 3
  cols: 3
  data: [1., 0., 0.,
         0., 1., 0.,
         0., 0., 1.]
projection_matrix:
  rows: 3
  cols: 4
  data: [600.07017,   0.     , 284.95284,   0.     ,
          0.     , 807.38252, 216.33741,   0.     ,
```

```
       0.    ,  0.    ,  1.    ,  0.    ]
```